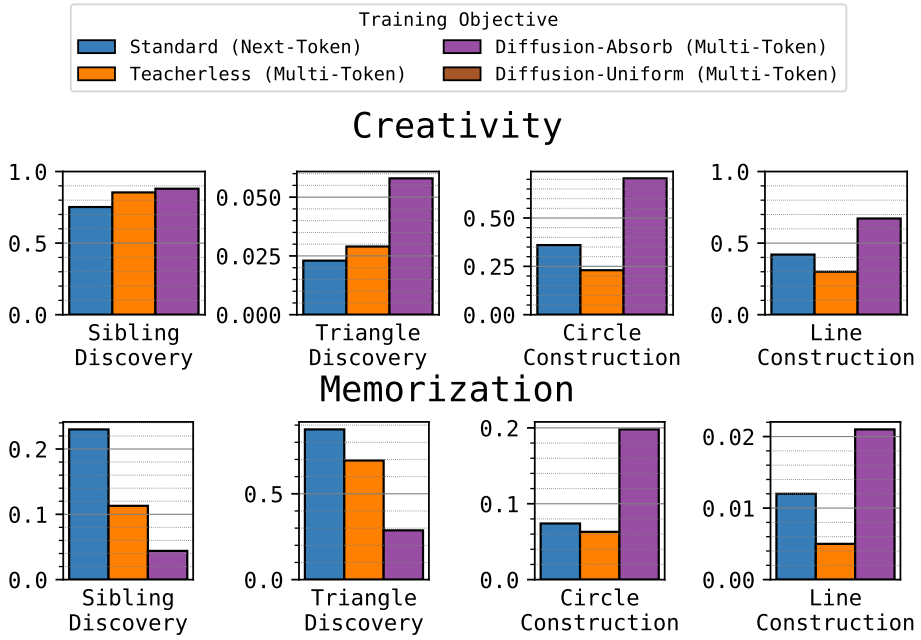

Figure 29. **On a medium-sized (∼400M) model, multi-token diffusion training improves algorithmic creativity from Eq 1 (top) on our four open-ended algorithmic tasks. We use greedy decoding in this plot.**

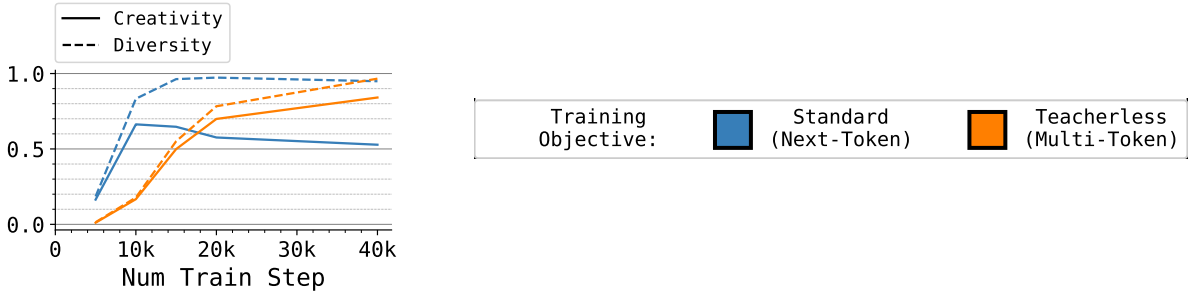

Figure 30. **Algorithmic creativity and diversity are not necessarily correlated, exhibiting distinct dynamics:** We find that NTP has a high diversity score through training, even higher than MTP. However, its algorithmic creativity reaches only a mediocre peak before descending, when MTP starts surpassing it. We use greedy decoding in this plot.

I. Experiments on summarization

Experimental Details. In Table 3, we provide the hyperparameter details for the GPT models finetuned on both XSUM (Narayan et al., 2018) and CNN/DailyMail (Nallapati et al., 2016) for one epoch. We use a learning rate with linear warm up for 0.05 of the total steps, followed by linear decay to 0. To measure Rouge and Self-Bleu, we generate and average across 5 summarizations per document, on a test dataset T of 250 datapoints. We finetune our models with either the NTP objective (Eq 2) or the teacherless MTP objective (Eq 3), with equal weight to both.

To measure quality, we compute the average of Rouge-1, Rouge-2, Rouge-L as Rouge. For measuring diversity, we generate five different summaries per test example, and compute Self-Bleu. This computes average pairwise sentence Bleu-2 scores with weights $(0.5, 0.5, 0, 0)$ on 1- and 2-tuples.

I.1. Additional graphs for effect of multi-token training

Fig 33 shows the diversity and quality graphs on the smaller-sized GPT-2 models on XSUM, and Fig 34 for CNN/DailyMail. While we consistently see improved quality from the multi-token model across the board, we don't see an increased diversity for fixed Rouge scores anymore.

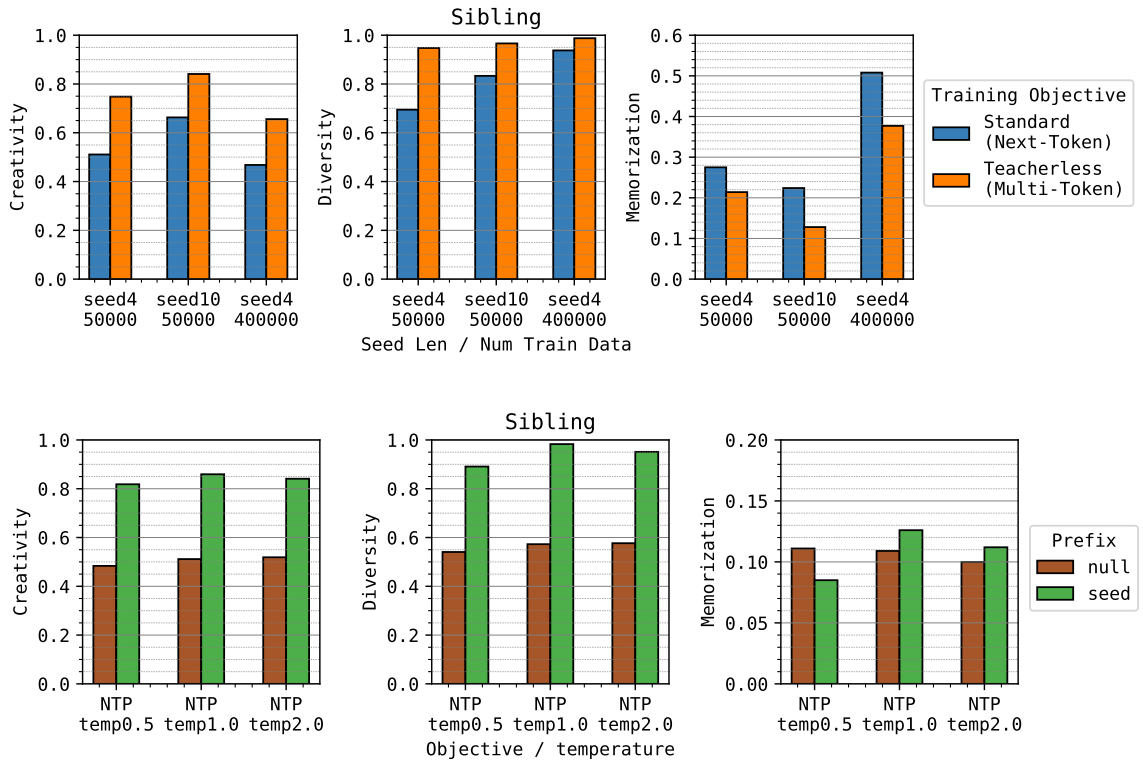

Figure 31. **Decomposition of algorithmic creativity for `GPT-2 (86M)` in `Sibling Discovery`:** We report algorithmic creativity, diversity and memorization at the checkpoint of best algorithmic creativity. We see that seed-conditioning contributes to higher diversity but does not help bring down memorization; teacherless training helps both diversity and in bringing down memorization.

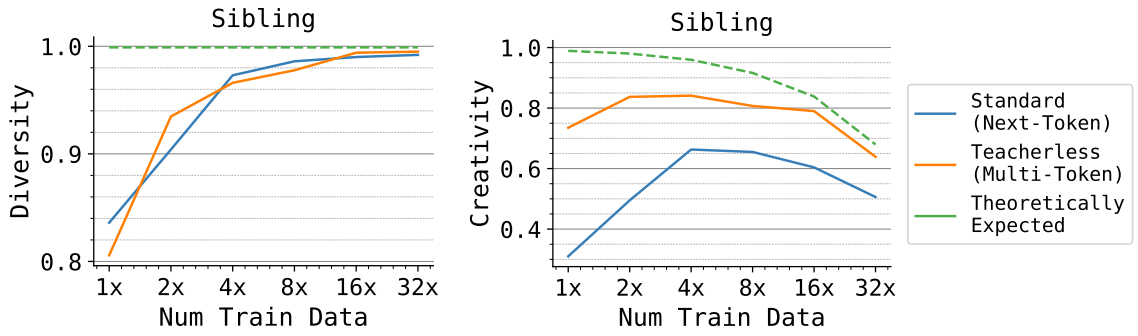

Figure 32. **Data scaling curve for algorithmic creativity and diversity:** As we increase the training data (for a fixed underlying graph), the theoretically expected maximum algorithmic creativity decreases as expected, while the theoretically expected maximum diversity stays the same. `NTP` tails to achieve the theoretically expected algorithmic creativity, while `MTP` almost achieves the theoretically expected performance at scale.

I.2. Effect of seed-conditioning

We also conducted seed-conditioning experiments as described in §3.1. The seed strings we use are 10 randomly sampled uppercase characters from the English alphabet. We report the quality-diversity plots in Fig 35 (for next-token prediction on `XSUM`) and Fig 36 (for multi-token prediction on `XSUM`). As such, we do not find any changes in diversity, perhaps because this is not a sufficiently open-ended task.

Table 3. **Hyperparameter details for summarization experiments.**

Hyperparameter	XSUM	CNN/DailyMail
Batch Size	32	32
Max. Learning Rate	5×10^{-5}	3×10^{-6}
Warmup Steps	338	124
Training Steps	7778	2486
Training Size	248906	79552

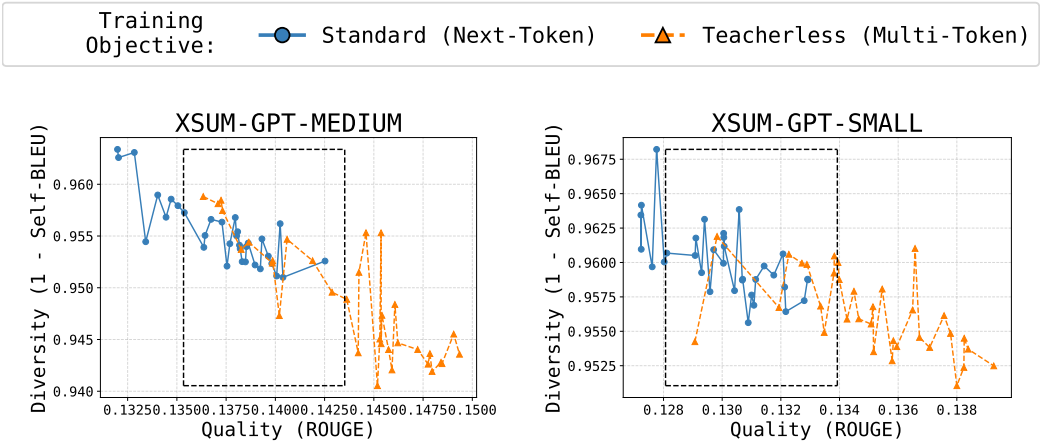

Figure 33. **Multi-Token Objective has no effect on diversity for smaller GPT models on XSUM.**

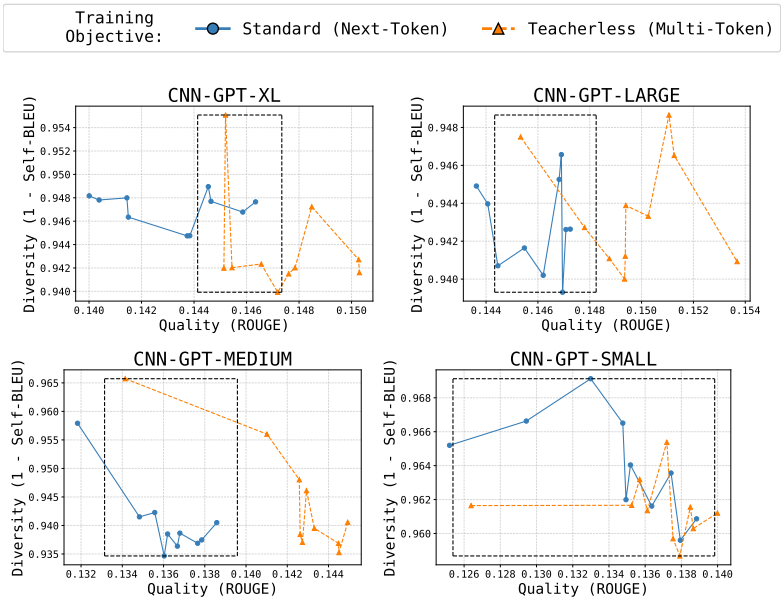

Figure 34. **Multi-Token Objective increases diversity for GPT-L and GPT-M but not for GPT-XL or GPT-S on CNN/DailyMail**